MIL-F-37782A
22 June 1978
SUPERSEDING
MIL-F-37782(DLA-DM)
25 April 1977

MILITARY SPECIFICATION

FLUOCINONIDE OINTMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Fluocinonide Ointment.

1.2 Classification. The ointment covered by this specification shall be of the following sizes (see 6.2):

Size 1 - 15 grams
Size 2 - 60 grams

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 So. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Fluocinonide Ointment, 0.05 percent.

3.1.1 Assay. The ointment shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the labeled amount of fluocinonide when assayed as specified in 4.7.1.

3.1.2 Identification. The chromatographic Rf values of the sample and reference standard obtained in 4.7.1 shall correspond.

3.2 Ingredients.

3.2.1 Fluocinonide. The fluocinonide used in the manufacture of the ointment shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

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3.3 Other ingredients. All other ingredients used in the manufacture of the ointment shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Expiration dating period. The expiration dating period shall be not less than 24 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

* 3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of ointment produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

* 4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample 2/
Assay	3.1.1	S-2	5 composite
Identification	3.1.2	S-2	5 composite

1/ Where possible, the sample shall be used for two or more tests.
2/ As an alternate, individual tests for assay and identification shall be performed on three (3) separate samples.

* 4.4 Examination. The finished ointment shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Ointment not free from foreign matter.
102	Ointment has crystal formation.
103	Ointment not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for fluocinonide.

Reagents and apparatus:

Fluocinonide reference standard solution - Weigh accurately about 25 mg of USP fluocinonide reference standard and transfer to a 25 ml volumetric flask. Dissolve and make to mark with chloroform.
Petroleum ether
60 percent methanol solution
Carbon tetrachloride
Solvent system for TLC - 75:25 by volume chloroform-acetone.
Glacial acetic acid

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Blue tetrazolium reagent - Dissolve 40 mg of BTZ in 50 ml of absolute ethanol.
Tetramethylammonium hydroxide reagent - Dilute 10 ml of a 10 percent aqueous TMAH solution to 100 ml in absolute ethanol.
Thin layer apparatus - "Desaga" or equivalent.
Spectrophotometer - Beckman DU or equivalent.

Procedure:

1. Add 75 ml of petroleum ether and 10 ml of the 60 percent methanol solution to a 125 ml separatory funnel.
2. Weigh accurately approximately 2 grams of the finished ointment and transfer it to the 125 ml separatory funnel.
3. Shake twice by hand, releasing the pressure each time, then shake on a mechanical shaker for 10 minutes or until the sample is completely dispersed. Allow to separate.
4. Drain the lower methanolic layer into a 250 ml separatory funnel containing 100 ml of purified water.
5. Re-extract the petroleum ether phase with additional 2x10 ml 60 percent methanol solution, shake each extract for 30 seconds, allow to separate. Drain and add the lower layer each time to the 250 ml separatory funnel. Discard the petroleum ether phase.
6. Stopper the 250 ml separatory funnel and shake well.
7. Extract the fluocinonide from the aqueous methanol solution with four 25 ml portions of carbon tetrachloride (CCL_4). (If emulsion occurs, 1 ml of a saturated sodium chloride solution may be added.). Filter each CCL_4 extract through a funnel, containing anhydrous sodium sulfate on a cotton pledget previously rinsed with CCL_4, into a 150 ml inverted conical flask. After draining the extracts through the sodium sulfate, rinse the funnel containing the sodium sulfate with additional CCL_4.
8. Evaporate the CCL_4 to dryness on a steam bath.
9. Dissolve the residue in chloroform and transfer quantitatively to a 1 ml volumetric flask. Make to the mark with chloroform, stopper and shake.

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10. Apply in band form a 50 µl aliquot of the sample solution from the 1 ml volumetric flask on a prepared silica plate. Using a separate section of the plate, apply in band form 50 µl of the fluocinonide reference standard solution.
11. Lower the plate into a clean chromatographic chamber lined with chromatographic paper saturated with the solvent system. Cover the tank with a lid and allow the solvent to ascend for a distance of 15 cm. Remove the plate and dry.
12. Examine the plate under 254 nm ultraviolet source and mark the areas encompassing the steroid spots. Scrape these areas and transfer to separate 30 ml test tubes. Treat an equal amount of silica gel adjacent to and at the same Rf value as the steroid spots for use as a reagent blank.
13. Add 5 ml of absolute ethanol to each tube and shake.
* 14. Separate the ethanol from the solids by centrifuging until a clear liquid is obtained.
* 15. Pipette 3 ml of the clear liquid into a third set of test tubes. Add 1 ml of blue tetrazolium reagent followed by the addition of 1 ml of 1 percent tetramethylammonium hydroxide reagent. Shake and store each tube in the dark for 60 minutes. At the end of this time period, add 1 ml of glacial acetic acid to each tube in order to quench the reaction. Mix well.
16. Determine the absorbances of each solution (standard, sample and reagent blank) against absolute ethanol on a suitable spectrophotometer at 525 nm. Subtract the absorbance of the reagent blank from those of the sample and standard, and use these net absorbances in the calculation.

* Calculation:

$$\frac{\text{Absorbance of Sample x Wt of Std (mg)}}{\text{Absorbance of Std x Wt of Sample(gram) x 25}} = \text{mg fluocinonide/gram}$$

$$\frac{\text{mg of fluocinonide/gram found x 100}}{\text{mg of fluocinolide/gram labeled}} = \text{percent of labeled amount}$$

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One tube containing 15 gram or 60 gram, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container: size 1 and size 2. The immediate container for size 1 and size 2 shall be a commercially available ointment tube considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twenty-four (24) units of issue of size 1 shall be packaged as specified in 5.2.12 of PPP-C-186, Twelve (12) units of issue of size 2 shall be packaged as specified in 5.2.13 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be Level A, B or C, as specified (see 6.1).

5.3.1 Level A. Five hundred seventy-six (576) units of issue of size 1 or one hundred forty-four (144) units of issue of size 2 shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Five hundred seventy-six (576) units of issue of size 1 or one hundred forty-four (144) units of issue of size 2 shall be packed as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

* 5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container: Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

* NOTES: (1) The National Stock Number need not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

* 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-00-355-7066	FLUOCINONIDE OINTMENT, 0.05%, 15 gram	Size 1
6505-01-022-2647	FLUOCINONIDE OINTMENT, 0.05%, 60 gram	Size 2

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DH

Project No. 6505-2021

Review activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-F-37782A FLUOCINONIDE OINTMENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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